

ENTERPRISE HEALTHCARE CREDENTIALING & LINKING SYSTEM

Master Technical, Functional, and Automation Specification

CONFIDENTIAL • AGES LEARNING SOLUTIONS | PROFICIO SPEECH | CHILD'S PLAY THERAPY

DOCUMENT METADATA & CONTROL

Project Reference ID: **AGES-SPEC-2026-V3**
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EXECUTIVE SCOPE & PURPOSE

This document establishes the authoritative, single-source specification for the clinical credentialing, payer enrollment, and facility group linking platform engineered for multi-entity pediatric therapy practices in California. It consolidates all requirements, business logic, multi-stage workflows, database schemas, UI component hierarchies, automation cadences, compliance reporting, and implementation milestones into one comprehensive reference.

Key Capabilities Covered in this Specification:

- 4-Step Clinician Intake, Verification, Payer Bundling, and Post-Add Provider Linking
- Collaborative Audit Trail & Contextual Notes System with Real-Time Author Attribution
- Supporting Document Verification Subtab with Cloud Storage Association & Status Tracking
- Dual-Verification Facility Group Linking Matrix (Legal Entity TIN + Location NPI)
- Real-Time Management Dashboard with 10 Executive KPIs and 30-120+ Day Aging Engine
- Section 5.11 Specialized Oversight Views (By Discipline, Payer, Specialist, Location)
- Section 5.11.6 Weekly (7 Core Metrics) and Section 5.11.7 Monthly (10 Sections) Reports
- Power BI DirectQuery Star-Schema Dataset Generator
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1.0 EXECUTIVE SUMMARY & ARCHITECTURAL VISION

1.1 Business Context & Problem Statement

Pediatric behavioral and rehabilitative therapy practices face intricate regulatory, clinical, and billing credentialing bottlenecks. A clinician cannot independently bill health plans until two mandatory criteria are fulfilled: (1) individual credentialing approval by the payer committee, and (2) legal facility group linking affiliating the provider's Type 1 individual NPI to the organization's Type 2 group NPI and physical clinic facility address.

Historically, credentialing operations across Ages Learning Solutions, Proficio Speech Therapy, and Child's Play Therapy suffered from fragmented spreadsheets, disconnected communication threads, untracked CAQH attestation expiration, and delayed facility roster additions. This system consolidates all credentialing workflows into an automated, transparent platform.

1.2 Core System Objectives & Strategic Goals

- **Accelerate Turnaround Time (TAT):** Reduce onboarding-to-billing lead times from 120+ days to under 45 days through structured validation gates and proactive follow-up alerts.
- **Eliminate Billing Claim Rejections:** Enforce dual verification (Payer Credentialing + Facility Linking) before billing release, preventing unlinked claim denials.
- **Unified Multi-Entity Transparency:** Provide real-time executive visibility across three distinct disciplines (ABA, Speech, OT) under a single consolidated reporting framework.
- **Collaborative Accountability:** Ensure seamless audit trails with immutable timestamps and user attribution for every status change, follow-up, and note.

2.0 MULTI-ENTITY GOVERNANCE & CLINICAL DISCIPLINES

2.1 Entity & Discipline Matrix

Entity Name	Discipline	Licensing Board / Body	Taxonomy Code	Primary Payers
Ages Learning Solutions	ABA	BACB (BCBA, BCaBA)	103K00000X / 106E00000X	Kaiser, Blue Shield, Optum, Aetna, Regional Ctrs
Proficio Speech Therapy	Speech (SLP)	CA SLPAHADB & ASHA	235Z00000X	Commercial Plans, Medi-Cal UMCs, School IEPs
Child's Play Therapy	OT	CBOT & NBCOT	225X00000X	Commercial PPO/HMO, Medi-Cal PAVE, Regional Ctrs

3.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

[FR-1] Multi-Step Clinician Intake & Credentialing Workflow

System must guide users through a guided 4-step wizard: Step 1 (Clinician Demographics & Discipline), Step 2 (NPI, State License, CAQH ID & Supporting Document Links), Step 3 (Payer Enrollment Bundling & Collaborative Comments), and Step 4 (Post-Add Summary & Facility Group Linking).

[FR-2] Provider Roster & Profile Management

Centralized repository of all clinical practitioners with automated NPI format verification, primary discipline tags, employment status tracking, and direct linkages to historical applications.

[FR-3] Multi-Payer Enrollment Tracking Pipeline

Track individual applications across standard lifecycle stages: Intake & Preparation, Submitted, Payer Review, Approved, and Linked/Effective. Support custom follow-up cadences and reference numbers.

[FR-4] Dual-Verification Facility & Group Linking Matrix

Grid interface mapping clinicians to physical facility locations and billing TINs. Prevent status marking as "Billing Released" until both Payer Approval and Group Linking are confirmed.

[FR-5] Collaborative Audit Trail & Real-Time Notes

Chronological timeline recording application milestones and user notes with author ID, author name, system role, and precise ISO timestamps. Notes must be non-prominent and accessible during workflow.

[FR-6] Supporting Document Links & Repository Subtab

Interactive subtab supporting document links (e.g. State License, Board Certification, Malpractice COI, CV, PAVE proof) with category tags, external cloud URLs, and expiration date tracking.

[FR-7] Management Dashboard & Aging Analysis Engine

Display 10 core operational KPIs (Total Providers, Total Apps, Submitted, Pending, Approved, Action Required, Overdue, Rejected, Avg Cycle Days, Linking Rate) and 30/60/90/120+ day aging analysis.

[FR-8] Section 5.11 Specialized Executive Oversight Views

Provide dedicated multi-tab views: 5.11.2 (By Discipline comparison), 5.11.3 (By Payer turnaround matrix), 5.11.4 (By Specialist workload distribution), and 5.11.5 (By Location clinic readiness roster).

[FR-9] Section 5.11.6 Weekly & 5.11.7 Monthly Reporting Suites

5.11.6 Weekly Report with 7 core metrics and escalation logs. 5.11.7 Monthly Report with 10 structured operational sections. Both must support print styling and CSV data export.

[FR-10] Power BI DirectQuery Star-Schema Model

Generate normalized Fact and Dimension tables (FactApplications, DimProviders, DimPayers, DimLocations, DimDates) with one-click JSON payload generation and REST endpoint compatibility.

4.0 END-TO-END OPERATIONAL LIFECYCLE WORKFLOWS

4.1 Step 1: Initial Clinician Onboarding Intake

Captures Field 1 (Full Name & Professional Credentials), Field 2 (Primary Phone & Work Email), Field 4 (Office Location / Service Hub), and Field 5 (Clinical Discipline: ABA, Speech, or OT). Includes pre-fill sample presets (BCBA, SLP, OT) for expedited test data entry.

4.2 Step 2: Provider Verification & Start Credentialing

Presents a summary banner of Step 1 details. Features a prominent "Start Credentialing" primary action. Collects Field 3 (Individual 10-digit NPI & State Professional License with Expiration) and Field 6 (CAQH ProView ID & Healthcare Taxonomy). Embeds the Supporting Document Links subtab allowing users to attach cloud document URLs.

4.3 Step 3: Credentialing Application Phase (Payers & Comments)

Presents an interactive checkbox grid of contracted insurance health plans (Kaiser, Blue Shield, Optum/UHC, Aetna, Cigna, Anthem, Magellan). Incorporates a non-prominent secondary "Add Comment" button that expands an inline note input field, storing audit comments with author name, role, and timestamp.

4.4 Step 4: Post-Add Confirmation & Billing Group Linking

Executes an atomic multi-collection persistence routine creating synchronized records across employees, clinical_staff, providers, applications, documents, and audit logs. Renders a comprehensive review card followed by a prominent Provider Linking section with a plus sign (+) button to assign group billing affiliations and clinic location NPIs.

4.5 Post-Approval Release & Re-Credentialing Cycles

Upon payer committee approval, the specialist logs the official Payer Effective Date. When both Payer Approval and Group Linking are established, the clinician transitions to "Linked / Billing Released". The system automatically initiates a 120-day CAQH attestation countdown and a 90-day license expiration reminder.

5.0 DATABASE ARCHITECTURE, SCHEMAS & DATA MODELS

5.1 Entity Relationship Overview

The system utilizes a relational-in-NoSQL Firestore structure ensuring referential integrity through explicit foreign key identifiers across six core collections:

- **employees:** Primary HR personnel entity (id, employeeNumber, firstName, lastName, email, phone, department, hireDate, status).
- **clinical_staff:** Clinical staff entity linked via employeeId (id, employeeId, discipline, credentials, npid, stateLicenseNumber, caqhId, locationId).
- **providers:** Provider roster entity linked via clinicalStaffId & employeeId (id, name, npid, caqhId, discipline, taxonomy, activeApplicationsCount).
- **applications:** Credentialing application instance (id, providerId, payerId, stage, submissionDate, approvalDate, effectiveDate, agingDays, isOverdue).
- **documents:** Verification document metadata (id, applicationId, providerId, title, category, fileUrl, expiryDate, status).
- **comments:** Collaborative audit log entry (id, applicationId, providerId, text, authorId, authorName, authorRole, timestamp).

5.2 Master Data Dictionary & Schema Definition

Field Name	Type	Collection	Constraints / Validation	Description
id	string	All	Regex: ^[a-zA-Z0-9_-]{1,64}\$	Immutable unique primary identifier
employeeId	string	clinical_staff	Required, references employees.id	Foreign key linking clinical profile to HR
clinicalStaffId	string	providers	Required, references clinical_staff.id	Foreign key linking provider to clinical staff
npid	string	clinical_staff	10-digit numeric format	National Provider Identifier (Type 1)
discipline	string	All	Enum: ABA Speech OT	Clinical healthcare discipline
stage	string	applications	Enum: 7 standard lifecycle stages	Current progress in credentialing pipeline
effectiveDate	string	applications	YYYY-MM-DD or null	Date health plan authorizes claims billing
fileUrl	string	documents	Valid URL (HTTPS recommended)	Cloud storage link to supporting credential

6.0 USER INTERFACE ARCHITECTURE & COMPONENT HIERARCHY

6.1 Design Principles & Anti-Slop Discipline

The user interface is engineered adhering strictly to enterprise healthcare software ergonomics: high contrast (slate/navy neutrals), deliberate 8px spatial grid, mathematical padding ratios (outer $\geq$ inner), zero artificial marketing hero blocks, zero low-contrast text, and 100% functional controls with responsive state feedback.

6.2 Primary View Hierarchy

- **Global Navigation Shell:** Header bar with quick switcher, notification bell, live Firestore sync indicator, and tab navigation.
- **Executive Dashboard (Tab 1):** 10 KPI cards, interactive 30-120+ day aging distribution, stage pipeline flow, and sub-views (5.11.2 - 5.11.5).
- **Applications Master Tracker (Tab 2):** Dual-view toggle: interactive Kanban drag-and-drop board + multi-column tabular grid with search, stage filters, and inline actions.
- **Facility Group Linking Matrix (Tab 3):** Roster view mapping providers to legal entities and physical practice locations with one-click link status toggles.
- **Clinical Staff Directory (Tab 4):** Card-based and table-based roster of all clinicians with credential expiration warnings and CAQH renewal flags.
- **Reports & Compliance Suite (Tab 5):** Weekly 5.11.6 and Monthly 5.11.7 reports with printable styling and Power BI star-schema JSON exporter.

7.0 BUSINESS LOGIC, CALCULATIONS & AUTOMATION ENGINE

7.1 Aging Bucketing & Time-in-Stage Algorithms

Application aging is calculated continuously against the initial submission timestamp:

- **0–30 Days (On Track):** Standard payer receipt and primary source verification window.
- **31–60 Days (In Progress):** Payer committee credentialing evaluation phase; routine follow-up cadence.
- **61–90 Days (Aging / Follow-up Due):** Exceeds standard commercial turnaround; automatic alert triggers for specialist outreach.
- **91–120 Days (Critical Attention):** Flagged as overdue; escalated in Weekly 5.11.6 report to Clinical Director.
- **120+ Days (Severe Delay / At-Risk):** Action required; payer moratorium or missing document verification flag.

7.2 CAQH ProView 120-Day Re-Attestation Countdown

The system monitors each clinician's last CAQH attestation date. At day 90 (30 days prior to expiration), an automated reminder badge appears on the provider's profile. At day 110 (10 days prior), high-priority escalation is logged to prevent CAQH profile lapse.

7.3 Medi-Cal PAVE Portal Stage Tracking

For Speech and Occupational Therapy clinicians billing Medi-Cal, the system enforces a strict 4-phase PAVE workflow: (1) PAVE Profile Creation & NPI Linking, (2) Board License Verification Upload, (3) Rendering Provider Agreement Signature, and (4) State Department of Health Care Services (DHCS) Approval Letter Receipt.

8.0 REPORTING, ANALYTICS & POWER BI INTEGRATION

8.1 Section 5.11.6 Weekly Operational Report

Every Monday morning, the system generates the 7 core executive metrics: (1) New applications initiated, (2) Applications submitted, (3) Scheduled follow-ups due, (4) Overdue applications (>90 days), (5) Additional documents requested by payers, (6) Approvals received, and (7) Active escalations with resolution notes. Fully formatted for browser print and CSV export.

8.2 Section 5.11.7 Monthly Credentialing Report

A comprehensive 10-section operational briefing: 1. Provider Updates, 2. Payer Policy Updates, 3. Applications Submitted, 4. Approvals Received, 5. Billing Effective Dates, 6. Pending Pipeline, 7. Operational Delays & Moratoriums, 8. Additional Documentation Requests, 9. Key Accomplishments, and 10. Upcoming Actions. Features tabbed views for Consolidated, ABA, Speech, and OT.

9.0 SECURITY, AUTHENTICATION & ACCESS CONTROL

9.1 Role-Based Access Control (RBAC)

Role	Dashboard & Roster	Create / Edit Apps	Facility Linking	Admin Settings
Administrator	Full Access	Full Access	Full Access	Full Access
Credentialing Specialist	Full Access	Full Access	Full Access	No Access
Credentialing Manager	Full Access	Full Access	Full Access	Audit Only
Read-Only Stakeholder	View Only	No Access	No Access	No Access

9.2 Hardened
Firestore Security
Rules

All Firestore collections (employees, clinical_staff, providers, applications, payers, facilities, documents, comments) enforce strict path variable regex validation: isValidId(docId) ensuring strings contain only alphanumeric characters, underscores, or hyphens (1-64 chars), preventing path traversal and injection attacks.

10.0 NON-FUNCTIONAL & OPERATIONAL REQUIREMENTS

- **System Latency:** Sub-200ms response time for in-memory KPI calculations, list filtering, and local state transitions.
- **Data Durability & Backup:** Multi-tier storage combining reactive state, browser persistence, formatted Excel/CSV export, and direct Firebase Firestore replication.
- **HIPAA Compliance:** No unencrypted transmission of Protected Health Information (PHI) or clinician PII. Secure token-based access.

11.0 IMPLEMENTATION ROADMAP & INTEGRATION BLUEPRINT

Phase / Sprint	Core Deliverables	Target System	Current Status
Phase 1 (Current)	Complete 4-Step Intake, Linking Matrix, Dashboard, 5.11 Suite, Rules	React 19 / Firestore	100% Operational
Phase 2 (Q3 2026)	Direct CAQH ProView B2B Gateway & Outbound Power Email Data	CAQH B2B API / SMTP	Architecture Staged
Phase 3 (2027)	Automated Medi-Cal PAVE RPA Submission Bot	Headless RPA Worker	Long-term Roadmap

12.0 MASTER VERIFICATION TRACEABILITY MATRIX

Req ID	Functional Scope	Validation Criteria	Audit Status
FR-1	4-Step Clinician Intake	Step 1-4 fields, sample presets, carry-forward	Verified / Pass
FR-2	Provider Roster Engine	NPI verification, discipline mapping, status flags	Verified / Pass
FR-3	Payer Tracking Pipeline	7 lifecycle stages, aging counters, milestone log	Verified / Pass
FR-4	Group Linking Matrix	Dual-verification TIN/NPI check before billing release	Verified / Pass
FR-5	Audit Trail & Notes	ISO timestamp, user ID, user name, system role	Verified / Pass
FR-6	Document Links Subtab	Cloud URL link, category tag, expiration tracking	Verified / Pass
FR-7	Executive Dashboard	10 KPIs, 30-120+ day aging, overdue queue	Verified / Pass
FR-8	5.11 Specialized Views	Discipline, Payer, Specialist, Location matrices	Verified / Pass
FR-9	5.11.6 / 5.11.7 Reports	Weekly 7 metrics, Monthly 10 sections, CSV & Print	Verified / Pass
FR-10	Power BI DirectQuery	Star-schema Fact/Dimension payload generator	Verified / Pass

